

Trial Examination 2024

HSC Year 12 Physics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Draw diagrams using pencil
- Calculators approved by NESA may be used
- A data sheet, formulae sheet, and Periodic Table are provided at the back of this paper

Total Marks: 100

Section I – 20 marks (pages 2–9)

- Attempt Questions 1–20
- Allow about 35 minutes for this section

Section II – 80 marks (pages 11–32)

- Attempt Questions 21–34
- Allow about 2 hours and 25 minutes for this section

Students are advised that this is a trial examination only and cannot in any way guarantee the content or the format of the 2024 HSC Year 12 Physics examination.

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SECTION I**20 marks****Attempt Questions 1–20****Allow about 35 minutes for this section**

Use the multiple-choice answer sheet for Questions 1–20.

- 1 Which of the following statements about the speed of light is correct?
- A. The speed of light in a vacuum is independent of the speed of the light source and the observer.
 - B. The speed of light in a vacuum decreases if the light source moves towards the observer.
 - C. The speed of light in a vacuum increases if the light source moves away from the observer.
 - D. The speed of light in a vacuum increases if the observer moves towards the light source.

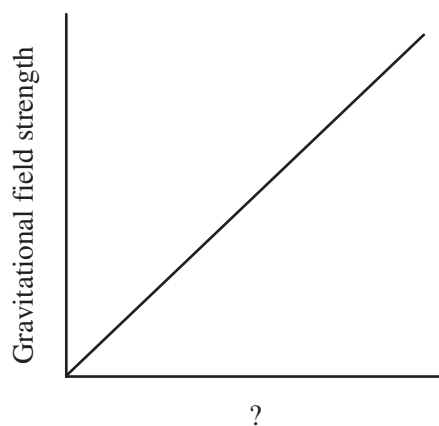
- 2 The table shows data about three stars, *X*, *Y* and *Z*, in the Milky Way.

<i>Star</i>	<i>Spectral class</i>	<i>Distance from the Sun (parsecs)</i>	<i>Apparent magnitude</i>
<i>X</i>	M2	194	+0.51
<i>Y</i>	B5	25	+0.57
<i>Z</i>	A2	439	+1.34

Which of the following statements is correct?

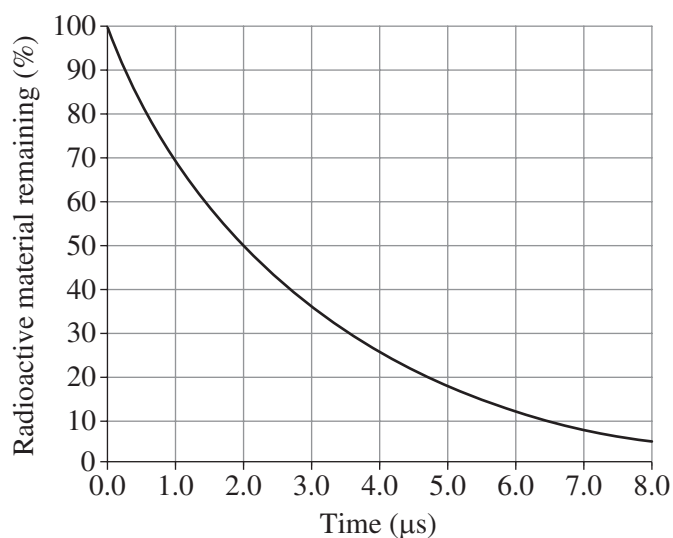
- A. Star *X* has the highest surface temperature.
 - B. Star *Y* has the highest surface temperature.
 - C. Stars *X* and *Y* have similar surface temperatures.
 - D. Star *Z* has the highest surface temperature.
- 3 During a game of paintball, player *X* shoots a paintball, which has a velocity of 80 m s^{-1} , horizontally from a height of 10 m towards player *Y*, who is positioned 25 m away. What is the horizontal velocity of the paintball just before impact?
- A. 8.0 m s^{-1}
 - B. 8.2 m s^{-1}
 - C. 80 m s^{-1}
 - D. 800 m s^{-1}

- 4 A Physics student drew the graph shown.



Which of the following could be an appropriate title for the x -axis?

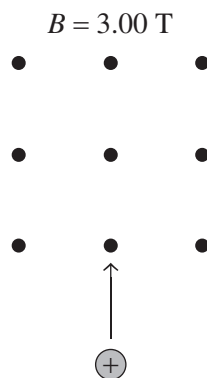
- A. distance from the centre of the planet
 - B. inverse square of the distance from the surface of the planet
 - C. square of the distance from the centre of the planet
 - D. mass of the planet
- 5 The graph shows the amount of a particular radioactive material that remains over a period of time.



Based on the graph, what is the half-life of the radioactive material?

- A. 2.0×10^{-6} s
- B. $1.0 \mu\text{s}$
- C. 2.0×10^{-3} s
- D. 2.0 s

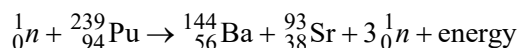
- 6 Which of the following describes Einstein's mass–energy equivalence relationship most accurately?
- A. Mass and energy are independent quantities.
 - B. The rest mass of an object is determined by its inertial frame of reference.
 - C. Mass and energy are identical.
 - D. Mass is a form of energy.
- 7 The gravitational force between two objects is F when the objects are separated by distance r . The distance between the objects is decreased by one third, and the mass of ONE of the objects is increased by a factor of 4. What is the representative force (F_1) between the objects?
- A. $F_1 = \frac{1}{36} F$
 - B. $F_1 = \frac{1}{3} F$
 - C. $F_1 = 12F$
 - D. $F_1 = 9F$
- 8 The diagram shows a proton being fired with a velocity of 1200 m s^{-1} into a uniform magnetic field of 3.00 T . The magnetic field is directed out of the page.



What is the acceleration of the proton as it enters the field?

- A. $5.78 \times 10^{-16} \text{ m s}^{-2}$ to the right
- B. $5.78 \times 10^{-16} \text{ m s}^{-2}$ to the left
- C. $3.45 \times 10^{11} \text{ m s}^{-2}$ to the right
- D. $3.45 \times 10^{11} \text{ m s}^{-2}$ to the left

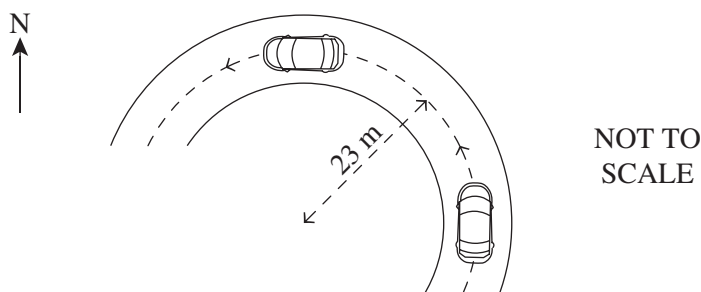
- 9 A plutonium-239 isotope is struck by a neutron; it absorbs the neutron and undergoes nuclear fission according to the following equation.



The combined mass of the plutonium-239 isotope's nucleus and the absorbed neutron is 3.99×10^{-25} kg. During the reaction, a mass of 3.07×10^{-28} kg is lost.

How much energy does the reaction release?

- A. 4.42×10^{-30} eV
 B. 3.99×10^{-25} eV
 C. 2.76×10^{-11} eV
 D. 1.72×10^8 eV
- 10 An instructor at a driving school demonstrated driving around a bend in a vehicle of mass 2200 kg, as shown in the diagram.



The instructor was initially driving north at a speed of 10.5 m s^{-1} and then maintained a constant speed of 10.5 m s^{-1} as they continued around the bend.

The magnitude of the frictional force between the road and the car's wheels, under different conditions, is shown in the table.

Condition	Frictional force (N)
dry	9500
heavy rain	8000
chemical spill on road (oily residue)	6500

Based on this information, under which conditions would it be safe to drive around the bend?

- A. none of the conditions
 B. dry only
 C. dry and heavy rain only
 D. all of the conditions

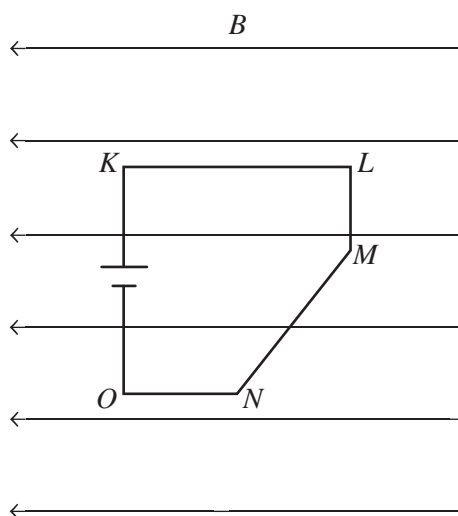
11 If bismuth-211 undergoes ONE alpha decay, what daughter nuclide is produced?

- A. $^{207}_{81}\text{Tl}$
 B. $^{207}_{85}\text{Tl}$
 C. $^{215}_{85}\text{At}$
 D. $^{215}_{87}\text{Fr}$

12 Which row of the table correctly compares the nucleosynthesis reactions that occur in main sequence stars?

	<i>Proton–proton chain</i>	<i>Carbon–nitrogen–oxygen (CNO) cycle</i>
A.	relatively slow and occurs in cooler main sequence stars	relatively fast and occurs in more massive, hotter main sequence stars
B.	relatively slow and occurs in more massive, hotter main sequence stars	relatively fast and occurs in cooler main sequence stars
C.	relatively fast and occurs in more massive, hotter main sequence stars	relatively slow and occurs in cooler main sequence stars
D.	relatively fast and occurs in cooler main sequence stars	relatively slow and occurs in more massive, hotter main sequence stars

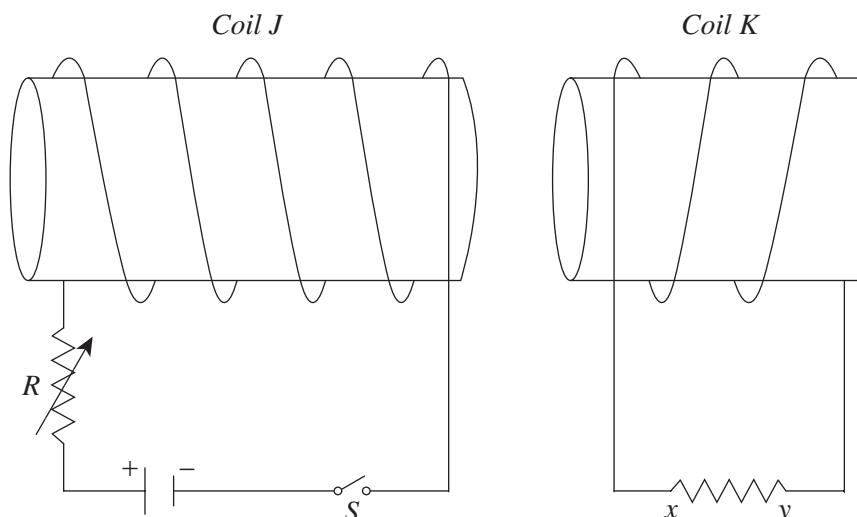
13 A current-carrying loop of wire, $KLMNO$, is placed in a uniform magnetic field that is directed to the left, as shown in the diagram.



Which row of the table describes the magnetic forces acting on sections LM , MN and NO of the loop?

	LM	MN	NO
A.	parallel out of the page	perpendicular into the page	perpendicular into the page
B.	perpendicular into the page	perpendicular out of the page	no magnetic force
C.	perpendicular into the page	perpendicular into the page	no magnetic force
D.	parallel into the page	parallel into the page	parallel into the page

- 14 The apparatus shown in the diagram is used in an experiment to demonstrate Lenz's law.



In the experiment, the following sequence of changes is made to the apparatus:

- I Switch S is opened after having been closed for a short period of time.
- II Coil K is brought closer to coil J and switch S is closed.
- III The resistance of R is decreased while switch S remains closed.

Which row of the table identifies the direction of the current in resistor xy after each change?

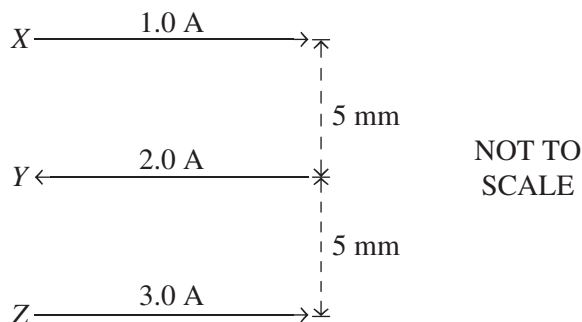
	<i>I</i>	<i>II</i>	<i>III</i>
A.	x to y	x to y	x to y
B.	x to y	y to x	y to x
C.	y to x	y to x	y to x
D.	y to x	x to y	y to x

- 15 The Moon is located 3.84×10^8 m from Earth and has a mass of 7.35×10^{22} kg. Earth has a mass of 6.0×10^{24} kg.

If a 1.00 kg object is located midway between Earth and the Moon, what is the object's acceleration due to gravity?

- A. $1.07 \times 10^{-2} \text{ m s}^{-2}$ towards Earth
- B. $1.07 \times 10^{-2} \text{ m s}^{-2}$ towards the Moon
- C. $1.08 \times 10^{-2} \text{ m s}^{-2}$ towards Earth
- D. $1.08 \times 10^{-2} \text{ m s}^{-2}$ towards the Moon

- 16 Three parallel current-carrying wires, X , Y and Z , are shown. The current in wire Y is flowing in the opposite direction to the currents in wires X and Z .

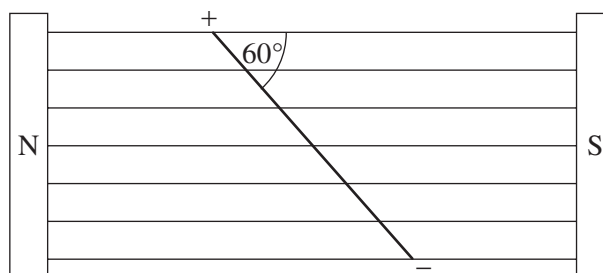


What is the magnitude of the net force acting on wire X ?

- A. $2 \times 10^{-5} \text{ N m}^{-1}$ up
 B. $2 \times 10^{-5} \text{ N m}^{-1}$ down
 C. $8 \times 10^{-5} \text{ N m}^{-1}$ up
 D. $8 \times 10^{-5} \text{ N m}^{-1}$ down
- 17 An observer on Earth claims to see an unidentified flying object (UFO). They estimate that the UFO has a length of 15.0 m and a width of 3.50 m, and is flying at a speed of $0.450 c$. What would the pilot of the UFO measure the UFO's length and width to be?

	<i>Length (m)</i>	<i>Width (m)</i>
A.	16.8	3.92
B.	13.4	3.50
C.	16.8	3.50
D.	13.4	3.92

- 18 A straight current-carrying conductor that has a length of 5.0 cm is suspended in a uniform magnetic field of 1.0 T at an angle of 60° to the field, as shown in the diagram.



If the current passing through the conductor is 2 A, what is the magnitude of the force acting on the conductor?

- A. 0.087 N into the page
 B. 0.087 N out of the page
 C. 8.7 N into the page
 D. 8.7 N out of the page

- 19** A satellite is currently in a low Earth orbit of 850 km and must be brought down to a lower orbit of 650 km.

In order to be stable in the lower orbit, the speed of the satellite would need to

- A. increase by 105 m s^{-1} .
- B. decrease by 105 m s^{-1} .
- C. increase by 3115 m s^{-1} .
- D. decrease by 3115 m s^{-1} .

- 20** A team of physicists are researching two planets, Endo and Exo, that orbit a distant star. The table shows some of the data that the team has gathered.

<i>Planet</i>	<i>Radius of orbit (m)</i>	<i>Orbital period (s)</i>
Endo	6.2×10^8	4.2×10^4
Exo	1.56×10^9	1.68×10^5

Based on the data in the table, which of the following statements is correct?

- A. The acceleration due to gravity experienced by Endo is 2.5 times greater than the acceleration due to gravity experienced by Exo.
- B. The acceleration due to gravity experienced by Exo is 2.5 times greater than the acceleration due to gravity experienced by Endo.
- C. The acceleration due to gravity experienced by Endo is 6.3 times greater than the acceleration due to gravity experienced by Exo.
- D. The acceleration due to gravity experienced by Exo is 6.3 times greater than the acceleration due to gravity experienced by Endo.

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HSC Year 12 Physics

Section II Answer Booklet

80 marks

Attempt Questions 21–34

Allow about 2 hours and 25 minutes for this section

Instructions

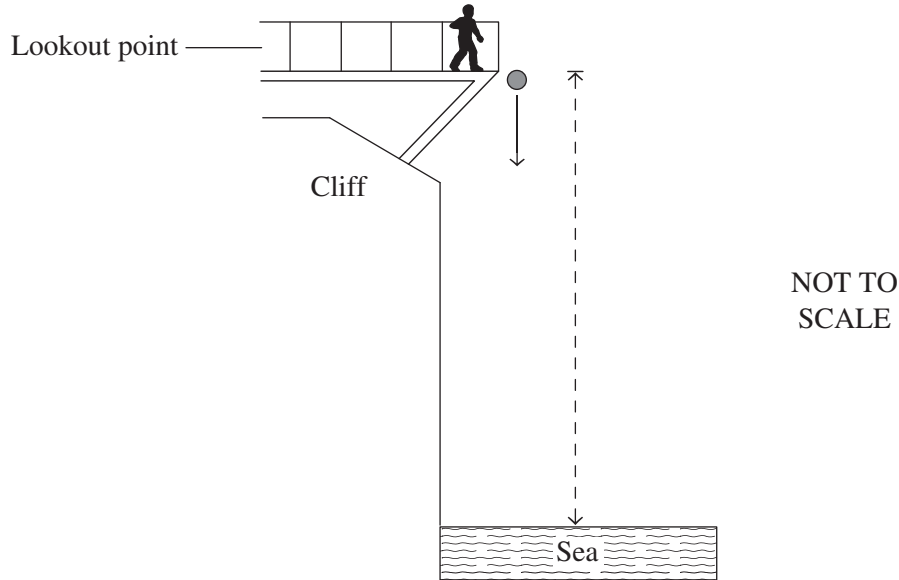
- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
 - Show all relevant working in questions involving calculations.
 - Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
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Please turn over

Question 21 (5 marks)

A popular tourist site has a lookout point at the top of a cliff overlooking the sea.

- (a) A visitor accidentally rolls a ball off the lookout point, as shown in the diagram. It takes 3.5 seconds for the ball to land in the sea below. **2**



Calculate the vertical distance that the ball travels before it lands in the sea.

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- (b) The tourist site also offers helicopter sightseeing tours. **3**
- A passenger in one of the helicopters drops their phone while trying to take a photo of the scenery. When the phone is dropped, the helicopter is 200 m above the ground and travelling at a horizontal speed of 50.0 m s^{-1} .

Calculate the horizontal distance that the phone travels before it lands on the ground.

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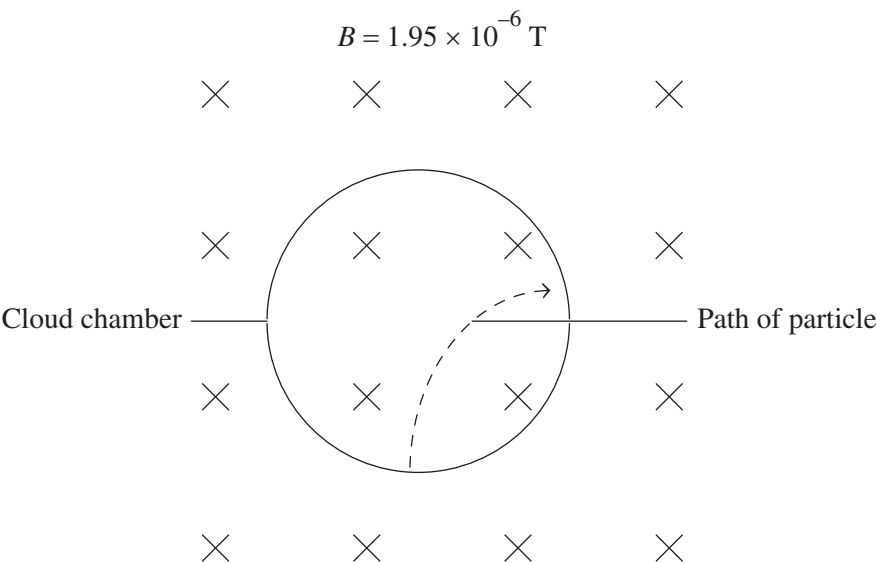
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Question 22 (4 marks)

A particle in an accelerator reaches a velocity of $2.50 \times 10^4 \text{ m s}^{-1}$ before entering a Wilson cloud chamber. The cloud chamber is in a uniform magnetic field of $1.95 \times 10^{-6} \text{ T}$. The path of the particle in the cloud chamber is illustrated in the diagram.



- (a) State whether the particle is positively or negatively charged. 1

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- (b) The radius of the particle's path is 5.0 cm. 3

Show that the mass-to-charge ratio of the particle is $2.6 \times 10^{11} \text{ C kg}^{-1}$.

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Question 23 (8 marks)

A Physics student is provided with two pairs of sunglasses and two light sources. One pair of sunglasses and one light source are polarised.

- (a) Explain how the student could identify which pair of sunglasses and which light source are polarised. **3**

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- (b) Discuss how polarisation can be explained by comparing the particle model and wave models of light. **5**

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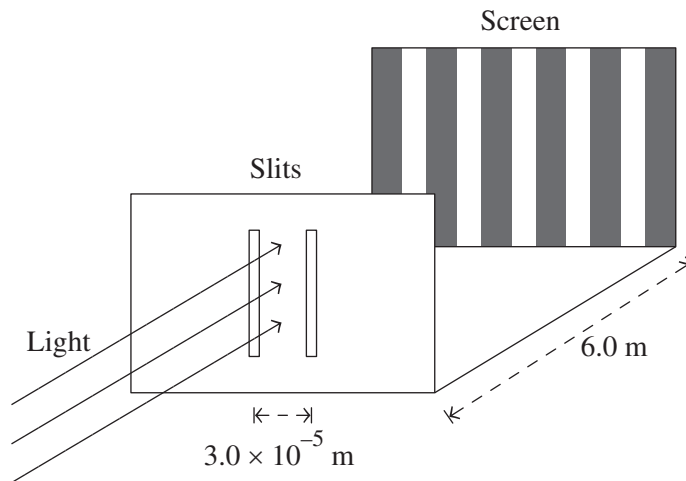
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Question 24 (4 marks)

The diagram illustrates an experiment where light of wavelength 500 nm is incident upon a pair of slits that are 3.0×10^{-5} m apart. A pattern of maxima and minima is produced on a screen that is 6.0 m behind the slits.



- (a) Calculate the distance between each of the maxima.

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- (b) Determine whether the experiment supports the particle or wave model of light.
Explain your answer.

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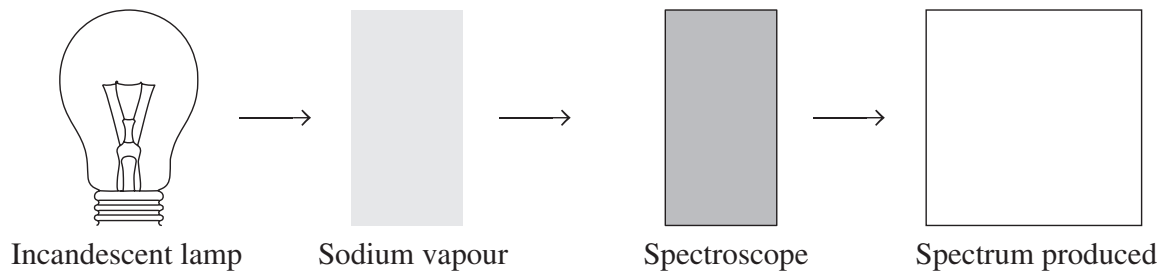
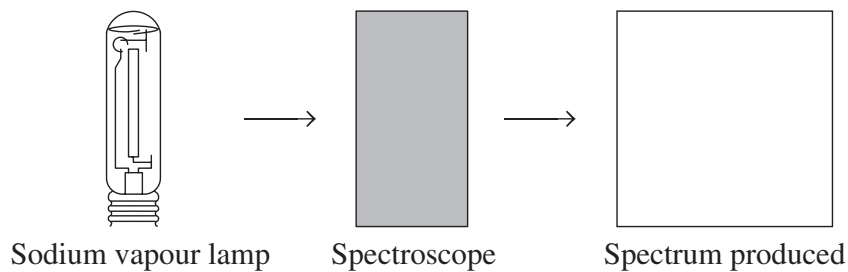
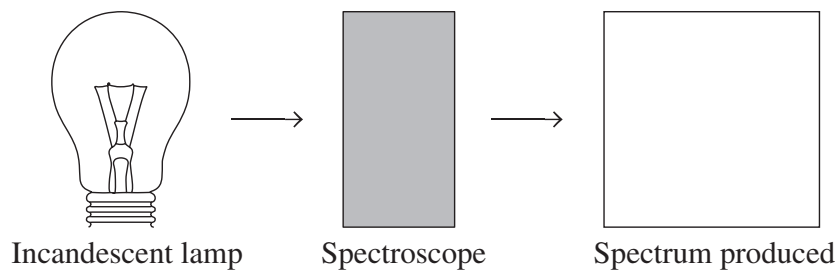
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Question 25 (7 marks)

A Physics class is investigating the spectra produced by three light sources, as shown in the diagram. They determine that each light source will produce one of the following spectra.

- spectrum I: two yellow lines on a black background
- spectrum II: the full range of colours with two black lines
- spectrum III: the full range of colours



Question 25 continues on page 17

Question 25 (continued)

Complete the diagram on page 16 by identifying the spectrum (I, II or III) that each light source will produce, and explain whether each spectrum is a continuous, absorption or emission spectrum.

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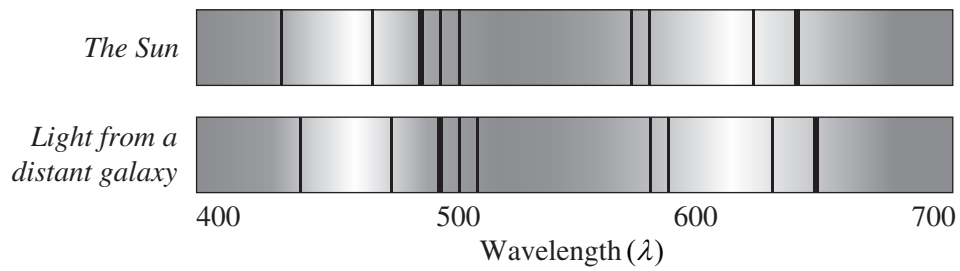
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End of Question 25

Question 26 (4 marks)

The spectra of the Sun and the light from a distant galaxy are shown.

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Explain how the spectra support Hubble's law and the Big Bang theory.

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Question 27 (5 marks)

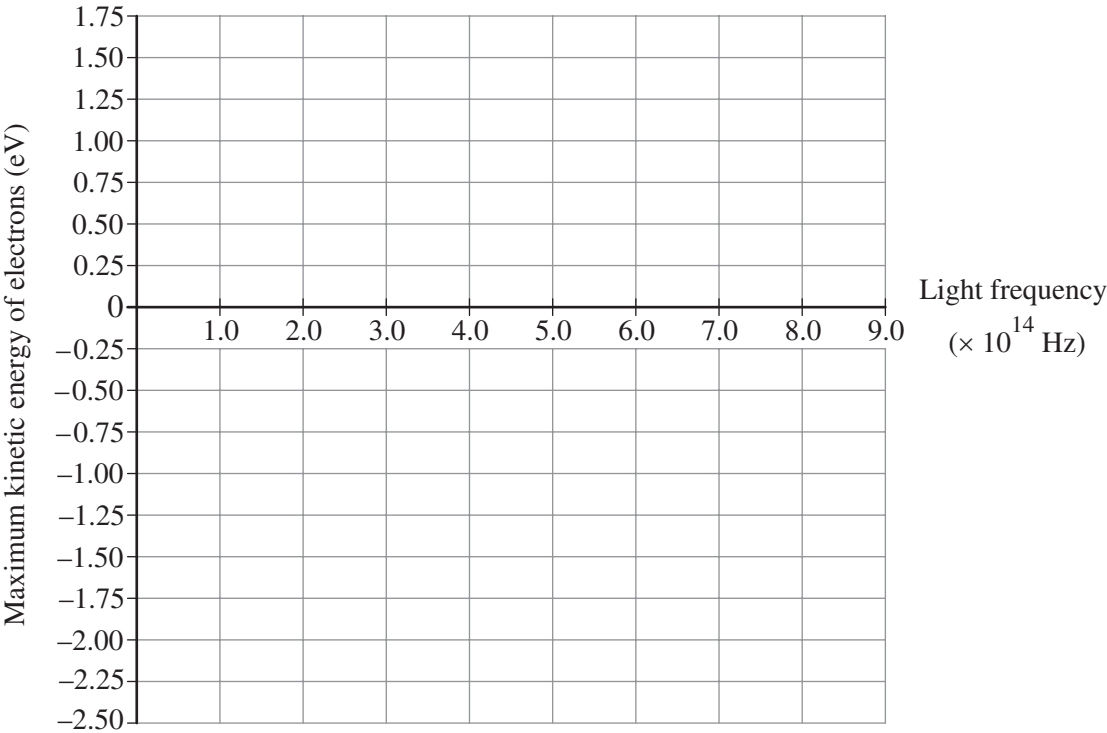
A team of physicists conducted an experiment to investigate the photoelectric effect. They directed light of different frequencies at the surface of a particular metal. The maximum kinetic energies of the electrons emitted by the metal were measured and the data was recorded in the table.

<i>Light frequency ($\times 10^{14}$ Hz)</i>	<i>Maximum kinetic energy of electrons (eV)</i>
5.6	0.01
6.0	0.18
6.5	0.41
7.1	0.70
7.9	0.97
8.8	1.31

- (a)

Plot the data on the axes provided and, using the plotted data:

3
- determine the work function (ϕ) of the metal
 - calculate a value for Planck’s constant (h).



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Question 27 continues on page 20

Question 27 (continued)

- (b) The team of physicists direct light of wavelength 420 nm at the metal. 2
Using the values for ϕ and h from part (a) on page 19, determine whether the photoelectric effect will occur.

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End of Question 27

Question 28 (4 marks)

A thought experiment proposes the following question.

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If I were sitting on a train that is travelling at the speed of light and I held a mirror up to my face, would I see my own reflection in the mirror?

There are two possible conclusions for this thought experiment.

- Conclusion I: Einstein would see his own reflection in the mirror.
- Conclusion II: Einstein would not see his own reflection in the mirror.

Discuss both possible conclusions for this thought experiment and determine which is correct.

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Question 29 (8 marks)

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Question 30 (6 marks)

A physicist conducted an experiment using a transformer. The results of the experiment are recorded in the table.

	<i>Primary coil</i>	<i>Secondary coil</i>
<i>Voltage (V)</i>	120	?
<i>Current (mA)</i>	3.0	30
<i>Number of turns</i>	?	5

- (a) Assuming that the transformer was 100% efficient, determine the secondary voltage and the number of turns in the primary coil.

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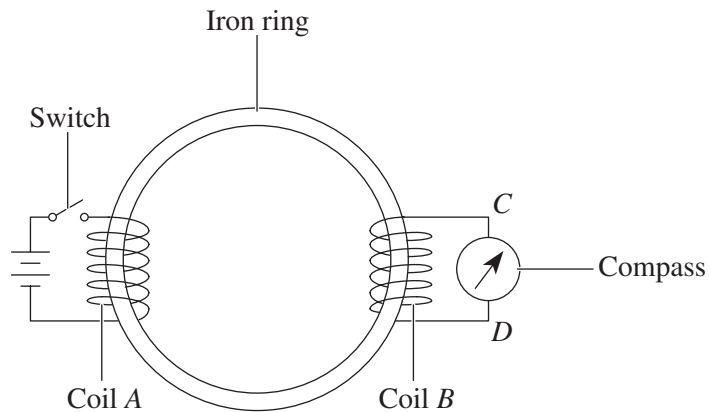
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Question 30 continues on page 24

Question 30 (continued)

- (b) Faraday's iron ring experiment used two coils and a soft iron ring, as shown in the diagram.

3

A compass was placed above the wire CD to detect any changes in the magnetic field produced. Faraday observed that the compass needle did not deflect when the switch was left open or closed, but it did deflect momentarily when the switch was initially opened or closed.

Explain Faraday's observations.

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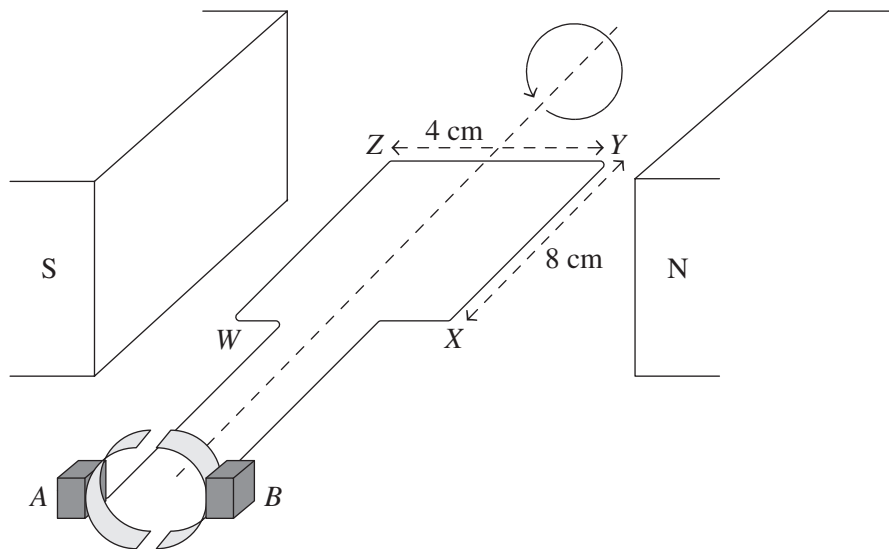
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End of Question 30

Question 31 (7 marks)

A simple DC motor is shown in the diagram.



- (a) On the diagram, label terminals *A* and *B* with their charges (+ or –). 1
- (b) One component of the motor ensures that the direction of the torque remains constant. 3
- With reference to the diagram, identify this component and explain why it is essential that the component maintains the direction of the torque.

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Question 31 continues on page 26

Question 31 (continued)

- (c) The magnets produce a magnetic field of 0.65 T. The magnetic field is at right angles to side XY , and a current of 12 A flows through loop $WXYZ$. **3**

The loop rotates from the position shown in the diagram on page 25 (position I), to a position where the plane of the loop makes an angle of 60° from the horizontal (position II), and then to a position where the plane of the loop makes an angle of 90° from the horizontal (position III).

Determine the magnitude of the torque and the direction of rotation at positions I, II and III.

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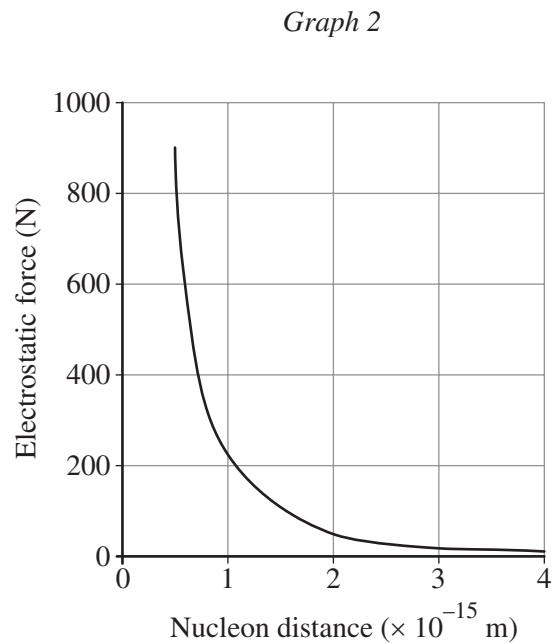
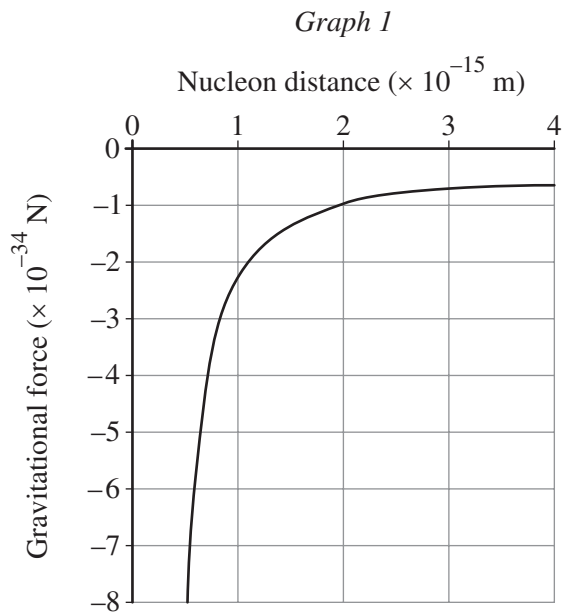
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End of Question 31

Question 32 (5 marks)

Graph 1 shows the relationship between gravitational force and nucleon distance, and graph 2 shows the relationship between electrostatic force and nucleon distance.

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Using the graphs and your knowledge of the fundamental forces, account for the stability of the nucleus.

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Question 33 (5 marks)

Outline how projectile motion could be modelled in a Physics classroom to show the relationship between launch angle and horizontal range. Assume that the projectile lands at the same height from which it was launched and support your answer with a graph.

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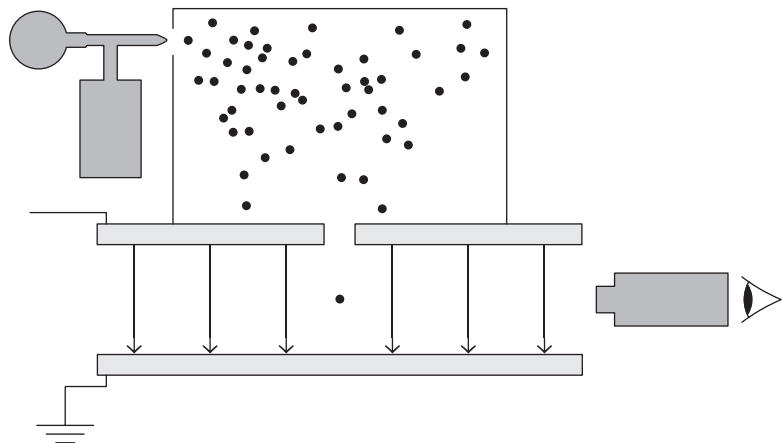
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Question 34 (8 marks)

(a) The diagram shows a simplified representation of Millikan’s oil drop experiment.

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Outline Millikan’s experiment AND how it provided evidence relating to the electron.

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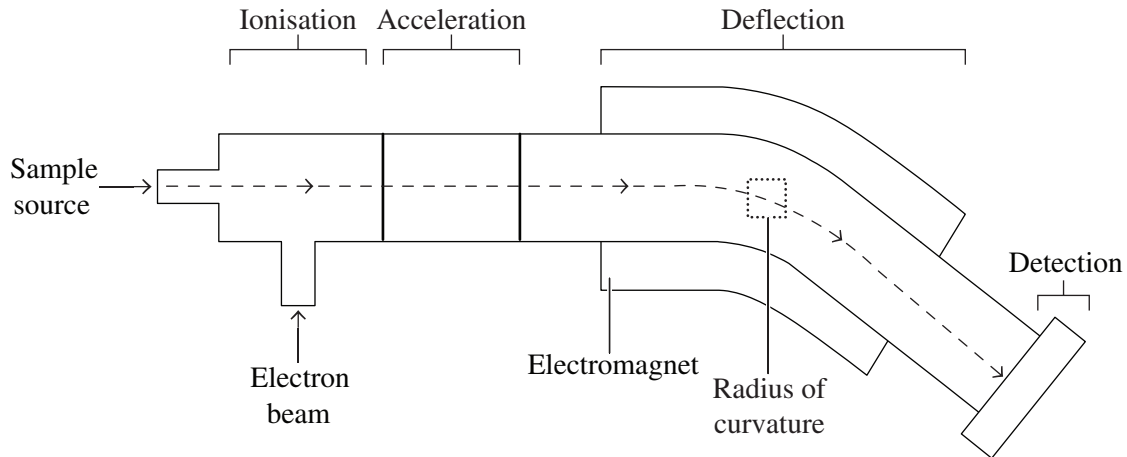
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Question 34 continues on page 30

Question 34 (continued)

- (b) Mass spectrometry is used to determine the mass of particles. A mass spectrometer projects a sample at a known velocity through an evacuated chamber. The sample is first bombarded with an electron beam, which causes particles in the sample to become positively charged ions with equal charges. The ions then enter a region of uniform magnetic field produced by an electromagnet that is perpendicular to the velocity of the ion. A simplified illustration of this process is shown.

3



Explain how the magnetic field enables the masses of the ions to be determined.

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End of paper

Section II extra writing space

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DATA SHEET

Charge on electron, q_e	$-1.602 \times 10^{-19} \text{ C}$
Mass of electron, m_e	$9.109 \times 10^{-31} \text{ kg}$
Mass of neutron, m_n	$1.675 \times 10^{-27} \text{ kg}$
Mass of proton, m_p	$1.673 \times 10^{-27} \text{ kg}$
Speed of sound in air	340 ms^{-1}
Earth's gravitational acceleration, g	9.8 ms^{-2}
Speed of light, c	$3.00 \times 10^8 \text{ ms}^{-1}$
Electric permittivity constant, ϵ_0	$8.854 \times 10^{-12} \text{ A}^2 \text{ s}^4 \text{ kg}^{-1} \text{ m}^{-3}$
Magnetic permeability constant, μ_0	$4\pi \times 10^{-7} \text{ N A}^{-2}$
Universal gravitational constant, G	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Mass of Earth, M_E	$6.0 \times 10^{24} \text{ kg}$
Radius of Earth, r_E	$6.371 \times 10^6 \text{ m}$
Planck constant, h	$6.626 \times 10^{-34} \text{ J s}$
Rydberg constant, R (hydrogen)	$1.097 \times 10^7 \text{ m}^{-1}$
Atomic mass unit, u	$1.661 \times 10^{-27} \text{ kg}$ $931.5 \text{ MeV}/c^2$
1 eV	$1.602 \times 10^{-19} \text{ J}$
Density of water, ρ	$1.00 \times 10^3 \text{ kg m}^{-3}$
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$
Wien's displacement constant, b	$2.898 \times 10^{-3} \text{ m K}$

FORMULAE SHEET

Motion, forces and gravity

$$s = ut + \frac{1}{2}at^2$$

$$v = u + at$$

$$v^2 = u^2 + 2as$$

$$\vec{F}_{\text{net}} = m\vec{a}$$

$$\Delta U = mg\Delta h$$

$$W = F_{\parallel}s = Fs \cos \theta$$

$$P = \frac{\Delta E}{\Delta t}$$

$$K = \frac{1}{2}mv^2$$

$$\sum \frac{1}{2}mv_{\text{before}}^2 = \sum \frac{1}{2}mv_{\text{after}}^2$$

$$P = F_{\parallel}v = Fv \cos \theta$$

$$\Delta \vec{p} = \vec{F}_{\text{net}}\Delta t$$

$$\sum m\vec{v}_{\text{before}} = \sum m\vec{v}_{\text{after}}$$

$$\omega = \frac{\Delta \theta}{t}$$

$$a_c = \frac{v^2}{r}$$

$$\tau = r_{\perp}F = rF \sin \theta$$

$$F_c = \frac{mv^2}{r}$$

$$v = \frac{2\pi r}{T}$$

$$F = \frac{GMm}{r^2}$$

$$U = -\frac{GMm}{r}$$

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

Waves and thermodynamics

$$v = f\lambda$$

$$f_{\text{beat}} = |f_2 - f_1|$$

$$f = \frac{1}{T}$$

$$f' = f \frac{(v_{\text{wave}} + v_{\text{observer}})}{(v_{\text{wave}} - v_{\text{source}})}$$

$$d \sin \theta = m\lambda$$

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$n_x = \frac{c}{v_x}$$

$$\sin \theta_c = \frac{n_2}{n_1}$$

$$I = I_{\text{max}} \cos^2 \theta$$

$$I_1 r_1^2 = I_2 r_2^2$$

$$Q = mc\Delta T$$

$$\frac{Q}{t} = \frac{kA\Delta T}{d}$$

FORMULAE SHEET (CONTINUED)**Electricity and magnetism**

$$E = \frac{V}{d}$$

$$\vec{F} = q\vec{E}$$

$$V = \frac{\Delta U}{q}$$

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

$$W = qV$$

$$I = \frac{q}{t}$$

$$W = qEd$$

$$V = IR$$

$$B = \frac{\mu_0 I}{2\pi r}$$

$$P = VI$$

$$B = \frac{\mu_0 NI}{L}$$

$$F = qv_{\perp} B = qvB \sin \theta$$

$$\Phi = B_{\parallel} A = BA \cos \theta$$

$$F = I l_{\perp} B = I l B \sin \theta$$

$$\mathcal{E} = -N \frac{\Delta \Phi}{\Delta t}$$

$$\frac{F}{l} = \frac{\mu_0}{2\pi} \frac{I_1 I_2}{r}$$

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

$$\tau = n l A_{\perp} B = n l A B \sin \theta$$

$$V_p I_p = V_s I_s$$

Quantum, special relativity and nuclear

$$\lambda = \frac{h}{mv}$$

$$t = \frac{t_0}{\sqrt{\left(1 - \frac{v^2}{c^2}\right)}}$$

$$K_{\max} = hf - \phi$$

$$l = l_0 \sqrt{\left(1 - \frac{v^2}{c^2}\right)}$$

$$\lambda_{\max} = \frac{b}{T}$$

$$p_v = \frac{m_0 v}{\sqrt{\left(1 - \frac{v^2}{c^2}\right)}}$$

$$E = mc^2$$

$$N_t = N_0 e^{-\lambda t}$$

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

$$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$$

PERIODIC TABLE OF THE ELEMENTS

PERIODIC TABLE OF THE ELEMENTS

1 H 1.008 hydrogen		KEY										2 He 4.003 helium																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
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3 Li 6.941 lithium	4 Be 9.012 beryllium	79 Au 197.0 gold	80 Hg 200.6 mercury	81 Tl 204.4 thallium	82 Pb 207.2 lead	83 Bi 209.0 bismuth	84 Po 209.0 polonium	85 At 209.0 astatine	86 Rn 222.0 radon	87 Fr 223.0 francium	88 Ra 226.0 radium	89-103 actinoids	90-103 transactinoids																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																			
11 Na 22.99 sodium	12 Mg 24.31 magnesium	21 Sc 44.96 scandium	22 Ti 47.87 titanium	23 V 50.94 vanadium	24 Cr 52.00 chromium	25 Mn 54.94 manganese	26 Fe 55.85 iron	27 Co 58.93 cobalt	28 Ni 58.69 nickel	29 Cu 63.55 copper	30 Zn 65.38 zinc	31 Ga 69.72 gallium	32 Ge 72.64 germanium	33 As 74.92 arsenic	34 Se 78.96 selenium	35 Br 79.90 bromine	36 Kr 83.80 krypton	37 Rb 85.47 rubidium	38 Sr 87.61 strontium	39 Y 88.91 yttrium	40 Zr 91.22 zirconium	41 Nb 92.91 niobium	42 Mo 95.96 molybdenum	43 Tc 98.0 technetium	44 Ru 101.1 ruthenium	45 Rh 102.9 rhodium	46 Pd 106.4 palladium	47 Ag 107.9 silver	48 Cd 112.4 cadmium	49 In 114.8 indium	50 Sn 118.7 tin	51 Sb 121.8 antimony	52 Te 127.6 tellurium	53 I 126.9 iodine	54 Xe 131.3 xenon	55 Cs 132.9 caesium	56 Ba 137.3 barium	57-71 lanthanoids	72 Hf 178.5 hafnium	73 Ta 180.9 tantalum	74 W 183.9 tungsten	75 Re 186.2 rhenium	76 Os 190.2 osmium	77 Ir 192.2 iridium	78 Pt 195.1 platinum	79 Au 197.0 gold	80 Hg 200.6 mercury	81 Tl 204.4 thallium	82 Pb 207.2 lead	83 Bi 209.0 bismuth	84 Po 209.0 polonium	85 At 209.0 astatine	86 Rn 222.0 radon	87 Fr 223.0 francium	88 Ra 226.0 radium	89-103 actinoids	90-103 transactinoids	91-103 transactinoids	94-103 superheavy elements	95-103 superheavy elements	96-103 superheavy elements	97-103 superheavy elements	98-103 superheavy elements	99-103 superheavy elements	100-103 superheavy elements	101-103 superheavy elements	102-103 superheavy elements	103-103 superheavy elements	104-103 superheavy elements	105-103 superheavy elements	106-103 superheavy elements	107-103 superheavy elements	108-103 superheavy elements	109-103 superheavy elements	110-103 superheavy elements	111-103 superheavy elements	112-103 superheavy elements	113-103 superheavy elements	114-103 superheavy elements	115-103 superheavy elements	116-103 superheavy elements	117-103 superheavy elements	118-103 superheavy elements	119-103 superheavy elements	120-103 superheavy elements	121-103 superheavy elements	122-103 superheavy elements	123-103 superheavy elements	124-103 superheavy elements	125-103 superheavy elements	126-103 superheavy elements	127-103 superheavy elements	128-103 superheavy elements	129-103 superheavy elements	130-103 superheavy elements	131-103 superheavy elements	132-103 superheavy elements	133-103 superheavy elements	134-103 superheavy elements	135-103 superheavy elements	136-103 superheavy elements	137-103 superheavy elements	138-103 superheavy elements	139-103 superheavy elements	140-103 superheavy elements	141-103 superheavy elements	142-103 superheavy elements	143-103 superheavy elements	144-103 superheavy elements	145-103 superheavy elements	146-103 superheavy elements	147-103 superheavy elements	148-103 superheavy elements	149-103 superheavy elements	150-103 superheavy elements	151-103 superheavy elements	152-103 superheavy elements	153-103 superheavy elements	154-103 superheavy elements	155-103 superheavy elements	156-103 superheavy elements	157-103 superheavy elements	158-103 superheavy elements	159-103 superheavy elements	160-103 superheavy elements	161-103 superheavy elements	162-103 superheavy elements	163-103 superheavy elements	164-103 superheavy elements	165-103 superheavy elements	166-103 superheavy elements	167-103 superheavy elements	168-103 superheavy elements	169-103 superheavy elements	170-103 superheavy elements	171-103 superheavy elements	172-103 superheavy elements	173-103 superheavy elements	174-103 superheavy elements	175-103 superheavy elements	176-103 superheavy elements	177-103 superheavy elements	178-103 superheavy elements	179-103 superheavy elements	180-103 superheavy elements	181-103 superheavy elements	182-103 superheavy elements	183-103 superheavy elements	184-103 superheavy elements	185-103 superheavy elements	186-103 superheavy elements	187-103 superheavy elements	188-103 superheavy elements	189-103 superheavy elements	190-103 superheavy elements	191-103 superheavy elements	192-103 superheavy elements	193-103 superheavy elements	194-103 superheavy elements	195-103 superheavy elements	196-103 superheavy elements	197-103 superheavy elements	198-103 superheavy elements	199-103 superheavy elements	200-103 superheavy elements	201-103 superheavy elements	202-103 superheavy elements	203-103 superheavy elements	204-103 superheavy elements	205-103 superheavy elements	206-103 superheavy elements	207-103 superheavy elements	208-103 superheavy elements	209-103 superheavy elements	210-103 superheavy elements	211-103 superheavy elements	212-103 superheavy elements	213-103 superheavy elements	214-103 superheavy elements	215-103 superheavy elements	216-103 superheavy elements	217-103 superheavy elements	218-103 superheavy elements	219-103 superheavy elements	220-103 superheavy elements	221-103 superheavy elements	222-103 superheavy elements	223-103 superheavy elements	224-103 superheavy elements	225-103 superheavy elements	226-103 superheavy elements	227-103 superheavy elements	228-103 superheavy elements	229-103 superheavy elements	230-103 superheavy elements	231-103 superheavy elements	232-103 superheavy elements	233-103 superheavy elements	234-103 superheavy elements	235-103 superheavy elements	236-103 superheavy elements	237-103 superheavy elements	238-103 superheavy elements	239-103 superheavy elements	240-103 superheavy elements	241-103 superheavy elements	242-103 superheavy elements	243-103 superheavy elements	244-103 superheavy elements	245-103 superheavy elements	246-103 superheavy elements	247-103 superheavy elements	248-103 superheavy elements	249-103 superheavy elements	250-103 superheavy elements	251-103 superheavy elements	252-103 superheavy elements	253-103 superheavy elements	254-103 superheavy elements	255-103 superheavy elements	256-103 superheavy elements	257-103 superheavy elements	258-103 superheavy elements	259-103 superheavy elements	260-103 superheavy elements	261-103 superheavy elements	262-103 superheavy elements	263-103 superheavy elements	264-103 superheavy elements	265-103 superheavy elements	266-103 superheavy elements	267-103 superheavy elements	268-103 superheavy elements	269-103 superheavy elements	270-103 superheavy elements	271-103 superheavy elements	272-103 superheavy elements	273-103 superheavy elements	274-103 superheavy elements	275-103 superheavy elements	276-103 superheavy elements	277-103 superheavy elements	278-103 superheavy elements	279-103 superheavy elements	280-103 superheavy elements	281-103 superheavy elements	282-103 superheavy elements	283-103 superheavy elements	284-103 superheavy elements	285-103 superheavy elements	286-103 superheavy elements	287-103 superheavy elements	288-103 superheavy elements	289-103 superheavy elements	290-103 superheavy elements	291-103 superheavy elements	292-103 superheavy elements	293-103 superheavy elements	294-103 superheavy elements	295-103 superheavy elements	296-103 superheavy elements	297-103 superheavy elements	298-103 superheavy elements	299-103 superheavy elements	300-103 superheavy elements	301-103 superheavy elements	302-103 superheavy elements	303-103 superheavy elements	304-103 superheavy elements	305-103 superheavy elements	306-103 superheavy elements	307-103 superheavy elements	308-103 superheavy elements	309-103 superheavy elements	310-103 superheavy elements	311-103 superheavy elements	312-103 superheavy elements	313-103 superheavy elements	314-103 superheavy elements	315-103 superheavy elements	316-103 superheavy elements	317-103 superheavy elements	318-103 superheavy elements	319-103 superheavy elements	320-103 superheavy elements	321-103 superheavy elements	322-103 superheavy elements	323-103 superheavy elements	324-103 superheavy elements	325-103 superheavy elements	326-103 superheavy elements	327-103 superheavy elements	328-103 superheavy elements	329-103 superheavy elements	330-103 superheavy elements	331-103 superheavy elements	332-103 superheavy elements	333-103 superheavy elements	334-103 superheavy elements	335-103 superheavy elements	336-103 superheavy elements	337-103 superheavy elements	338-103 superheavy elements	339-103 superheavy elements	340-103 superheavy elements	341-103 superheavy elements	342-103 superheavy elements	343-103 superheavy elements	344-103 superheavy elements	345-103 superheavy elements	346-103 superheavy elements	347-103 superheavy elements	348-103 superheavy elements	349-103 superheavy elements	350-103 superheavy elements	351-103 superheavy elements	352-103 superheavy elements	353-103 superheavy elements	354-103 superheavy elements	355-103 superheavy elements	356-103 superheavy elements	357-103 superheavy elements	358-103 superheavy elements	359-103 superheavy elements	360-103 superheavy elements	361-103 superheavy elements	362-103 superheavy elements	363-103 superheavy elements	364-103 superheavy elements	365-103 superheavy elements	366-103 superheavy elements	367-103 superheavy elements	368-103 superheavy elements	369-103 superheavy elements	370-103 superheavy elements	371-103 superheavy elements	372-103 superheavy elements	373-103 superheavy elements	374-103 superheavy elements	375-103 superheavy elements	376-103 superheavy elements	377-103 superheavy elements	378-103 superheavy elements	379-103 superheavy elements	380-103 superheavy elements	381-103 superheavy elements	382-103 superheavy elements	383-103 superheavy elements	384-103 superheavy elements	385-103 superheavy elements	386-103 superheavy elements	387-103 superheavy elements	388-103 superheavy elements	389-103 superheavy elements	390-103 superheavy elements	391-103 superheavy elements	392-103 superheavy elements	393-103 superheavy elements	394-103 superheavy elements	395-103 superheavy elements	396-103 superheavy elements	397-103 superheavy elements	398-103 superheavy elements	399-103 superheavy elements	400-103 superheavy elements	401-103 superheavy elements	402-103 superheavy elements	403-103 superheavy elements	404-103 superheavy elements	405-103 superheavy elements	406-103 superheavy elements	407-103 superheavy elements	408-103 superheavy elements	409-103 superheavy elements	410-103 superheavy elements	411-103 superheavy elements	412-103 superheavy elements	413-103 superheavy elements	414-103 superheavy elements	415-103 superheavy elements	416-103 superheavy elements	417-103 superheavy elements	418-103 superheavy elements	419-103 superheavy elements	420-103 superheavy elements	421-103 superheavy elements	422-103 superheavy elements	423-103 superheavy elements	424-103 superheavy elements	425-103 superheavy elements	426-103 superheavy elements	427-103 superheavy elements	428-103 superheavy elements	429-103 superheavy elements	430-103 superheavy elements	431-103 superheavy elements	432-103 superheavy elements	433-103 superheavy elements	434-103 superheavy elements	435-103 superheavy elements	436-103 superheavy elements	437-103 superheavy elements	438-103 superheavy elements	439-103 superheavy elements	440-103 superheavy elements	441-103 superheavy elements	442-103 superheavy elements	443-103 superheavy elements	444-103 superheavy elements	445-103 superheavy elements	446-103 superheavy elements	447-103 superheavy elements	448-103 superheavy elements	449-103 superheavy elements	450-103 superheavy elements	451-103 superheavy elements	452-103 superheavy elements	453-103 superheavy elements	454-103 superheavy elements	455-103 superheavy elements	456-103 superheavy elements	457-103 superheavy elements	458-103 superheavy elements	459-103 superheavy elements	460-103 superheavy elements	461-103 superheavy elements	462-103 superheavy elements	463-103 superheavy elements	464-103 superheavy elements	465-103 superheavy elements	466-103 superheavy elements	467-103 superheavy elements	468-103 superheavy elements	469-103 superheavy elements	470-103 superheavy elements	471-103 superheavy elements	472-103 superheavy elements	473-103 superheavy elements	474-103 superheavy elements	475-103 superheavy elements	476-103 superheavy elements	477-103 superheavy elements	478-103 superheavy elements	479-103 superheavy elements	480-103 superheavy elements	481-103 superheavy elements	482-103 superheavy elements	483-103 superheavy elements	484-103 superheavy elements	485-103 superheavy elements	486-103 superheavy elements	487-103 superheavy elements	488-103 superheavy elements	489-103 superheavy elements	490-103 superheavy elements	491-103 superheavy elements	492-103 superheavy elements	493-103 superheavy elements	494-103 superheavy elements	495-103 superheavy elements	496-103 superheavy elements	497-103 superheavy elements	498-103 superheavy elements	499-103 superheavy elements	500-103 superheavy elements	501-103 superheavy elements	502-103 superheavy elements	503-103 superheavy elements	504-103 superheavy elements	505-103 superheavy elements	506-103 superheavy elements	507-103 superheavy elements	508-103 superheavy elements	509-103 superheavy elements	510-103 superheavy elements	511-103 superheavy elements	512-103 superheavy elements	513-103 superheavy elements	514-103 superheavy elements	515-103 superheavy elements	516-103 superheavy elements	517-103 superheavy elements	518-103 superheavy elements	519-103 superheavy elements	520-103 superheavy elements	521-103 superheavy elements	522-103 superheavy elements	523-103 superheavy elements	524-103 superheavy elements	525-103 superheavy elements	526-103 superheavy elements	527-103 superheavy elements	528-103 superheavy elements	529-103 superheavy elements	530-103 superheavy elements	531-103 superheavy elements	532-103 superheavy elements	533-103 superheavy elements	534-103 superheavy elements	535-103 superheavy elements	536-103 superheavy elements	537-103 superheavy elements	538-103 superheavy elements	539-103 superheavy elements	540-103 superheavy elements	541-103 superheavy elements	542-103 superheavy elements	543-103 superheavy elements	544-103 superheavy elements	545-103 superheavy elements	546-103 superheavy elements	547-103 superheavy elements	548-103 superheavy elements	549-103 superheavy elements	550-103 superheavy elements	551-103 superheavy elements	552-103 superheavy elements	553-103 superheavy elements	554-103 superheavy elements	555-103 superheavy elements	556-103 superheavy elements	557-103 superheavy elements	558-103 superheavy elements	559-103 superheavy elements	560-103 superheavy elements	561-103 superheavy elements	562-103 superheavy elements	563-103 superheavy elements	564-103 superheavy elements	565-103 superheavy elements	566-103 superheavy elements	567-103 superheavy elements	568-103 superheavy elements	569-103 superheavy elements	570-103 superheavy elements	571-103 superheavy elements	572-103 superheavy elements	573-103 superheavy elements	574-103 superheavy elements	575-103 superheavy elements	576-103 superheavy elements	577-103 superheavy elements	578-103 superheavy elements	579-103 superheavy elements	580-103 superheavy elements	581-103 superheavy elements	582-103 superheavy elements	583-103 superheavy elements	584-103 superheavy elements	585-103 superheavy elements	586-103 superheavy elements	587-103 superheavy elements	588-103 superheavy elements	589-103 superheavy elements	590-103 superheavy elements	591-103 superheavy elements	592-103 superheavy elements	593-103 superheavy elements	594-103 superheavy elements	595-103 superheavy elements	596-103 superheavy elements	597-103 superheavy elements	598-103 superheavy elements	599-103 superheavy elements	600-103 superheavy elements	601-103 superheavy elements	602-103 superheavy elements	603-103 superheavy elements	604-103 superheavy elements	605-103 superheavy elements	606-103 superheavy elements	607-103 superheavy elements	608-103 superheavy elements	609-103 superheavy elements	610-103 superheavy elements	611-103 superheavy elements	612-103 superheavy elements	613-103 superheavy elements	614-103 superheavy elements	615-103 superheavy elements	616-103 superheavy elements	617-103 superheavy elements	618-103 superheavy elements	619-103 superheavy elements	620-103 superheavy elements	621-103 superheavy elements	622-103 superheavy elements	623-103 superheavy elements	624-103 superheavy elements	625-103 superheavy elements	626-103 superheavy elements	627-103 superheavy elements	628-103 superheavy elements	629-103 superheavy elements	630-103 superheavy elements	631-103 superheavy elements	632-103 superheavy elements	633-103 superheavy elements	634-103 superheavy elements	635-103 superheavy elements	636-103 superheavy elements	637-103 superheavy elements	638-103 superheavy elements	639-103 superheavy elements	640-103 superheavy elements	641-103 superheavy elements	642-103 superheavy elements	643-103 superheavy elements	644-103 superheavy elements	645-103 superheavy elements	646-103 superheavy elements	647-103 superheavy elements	648-103 superheavy elements	649-103 superheavy elements	650-103 superheavy elements	651-103 superheavy elements	652-103 superheavy elements	653-103 superheavy elements	654-103 superheavy elements	655-103 superheavy 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superheavy elements	692-103 superheavy elements	693-103 superheavy elements	694-103 superheavy elements	695-103 superheavy elements	696-103 superheavy elements	697-103 superheavy elements	698-103 superheavy elements	699-103 superheavy elements	700-103 superheavy elements	701-103 superheavy elements	702-103 superheavy elements	703-103 superheavy elements	704-103 superheavy elements	705-103 superheavy elements	706-103 superheavy elements	707-103 superheavy elements	708-103 superheavy elements	709-103 superheavy elements	710-103 superheavy elements	711-103 superheavy elements	712-103 superheavy elements	713-103 superheavy elements	714-103 superheavy elements	715-103 superheavy elements	716-103 superheavy elements	717-103 superheavy elements	718-103 superheavy elements	719-103 superheavy elements	720-103 superheavy elements	721-103 superheavy elements	722-103 superheavy elements	723-103 superheavy

Lanthanoids

57	La	138.9	lanthanum
59	Pr	140.9	praseodymium
61	Pm		promethium
63	Eu	152.0	europium
65	Tb	158.9	terbium
67	Ho	164.9	holmium
69	Tm	168.9	thulium
71	Lu	175.0	lutetium

Actinoids

89	Ac		actinium
91	Pa	231.0	protactinium
93	Np		neptunium
95	Am		americium
97	Bk		berkelium
99	Es		einsteinium
101	Md		mendelevium
103	Lr		lawrencium

Standard atomic weights are abridged to four significant figures.
Elements with no reported values in the table have no stable nuclides.
Information on elements with atomic numbers 113 and above is sourced from the International Union of Pure and Applied Chemistry Periodic Table of the Elements (November 2016 version).
The International Union of Pure and Applied Chemistry Periodic Table of the Elements (February 2010 version) is the principal source of all other data. Some data may have been modified.